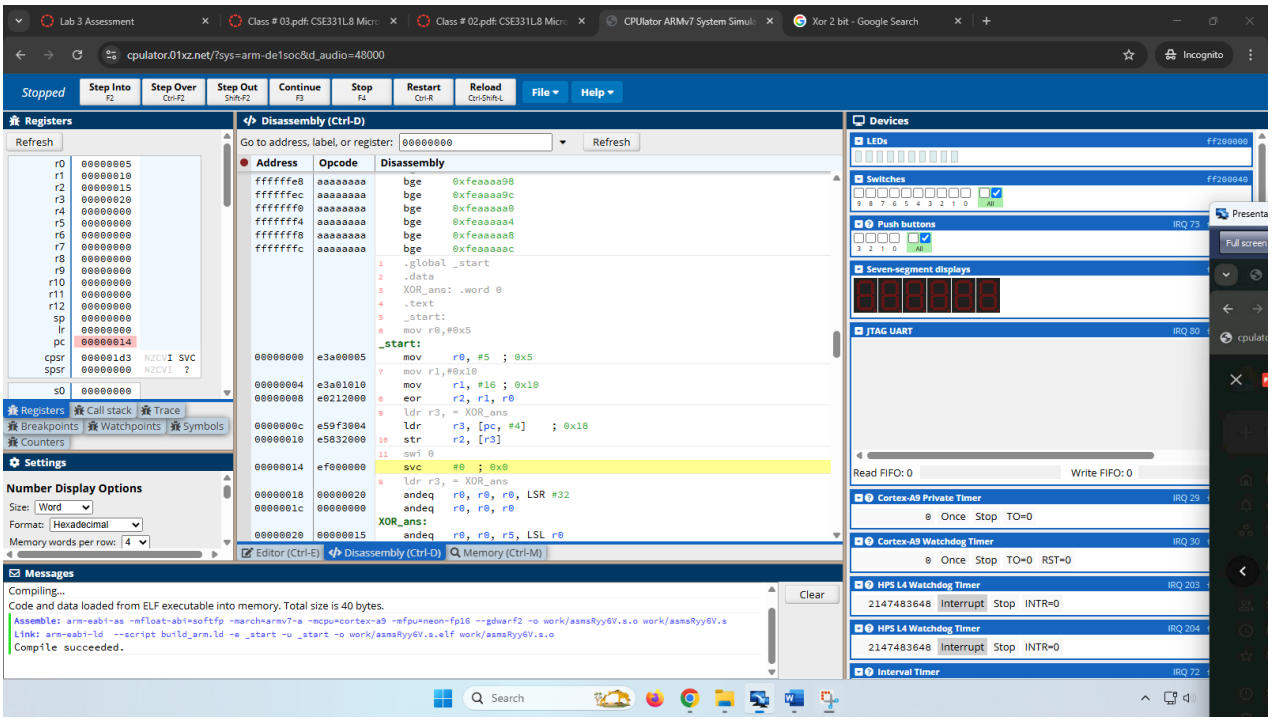
Name: MD. Rokib Hasan Oli

ID: 2211950642

Lab 3 Assessment

Q1

```
.global _start
.data
  XOR_ans: .word 0
.text
_start:
        mov r0,#0x5
        mov r1,#0x10
        eor r2,r1,r0
        ldr r3, = XOR_ans
        str r2,[r3]
        swi 0
```

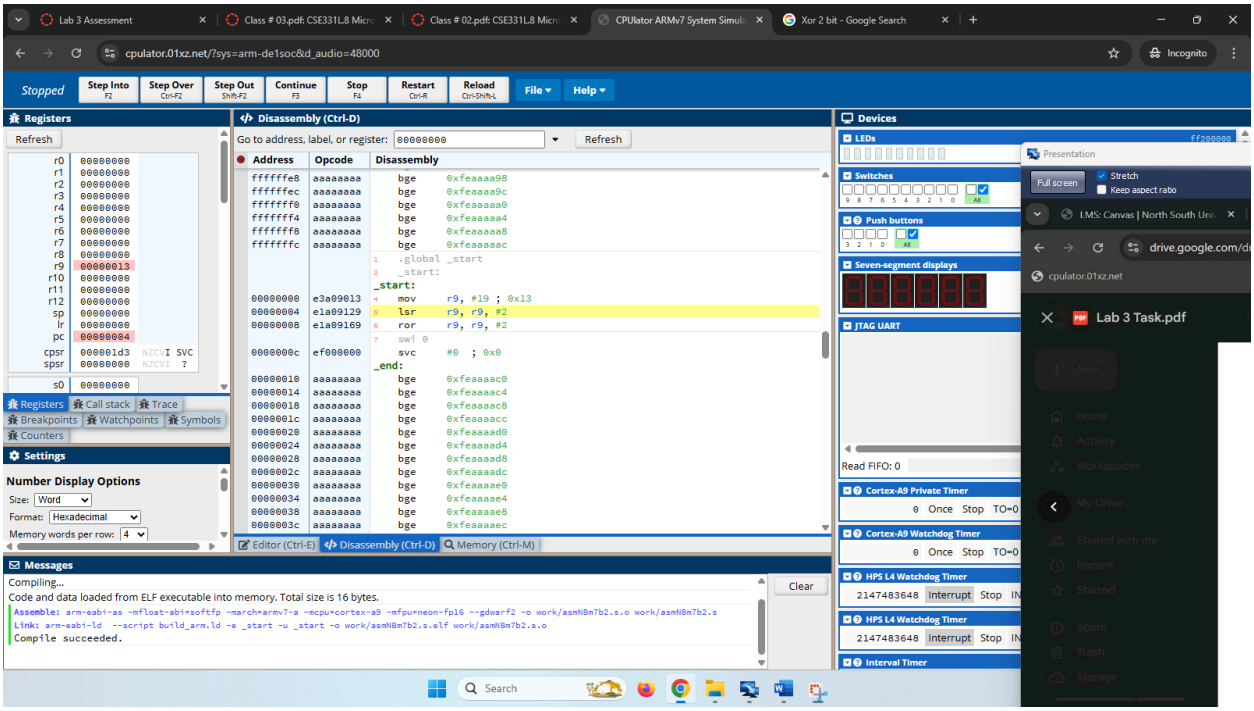


Q2

```
.global _start


_start:
        mov r9,#19
        lsr r9,r9,#2
        ror r9,r9,#2
        swi 0
```

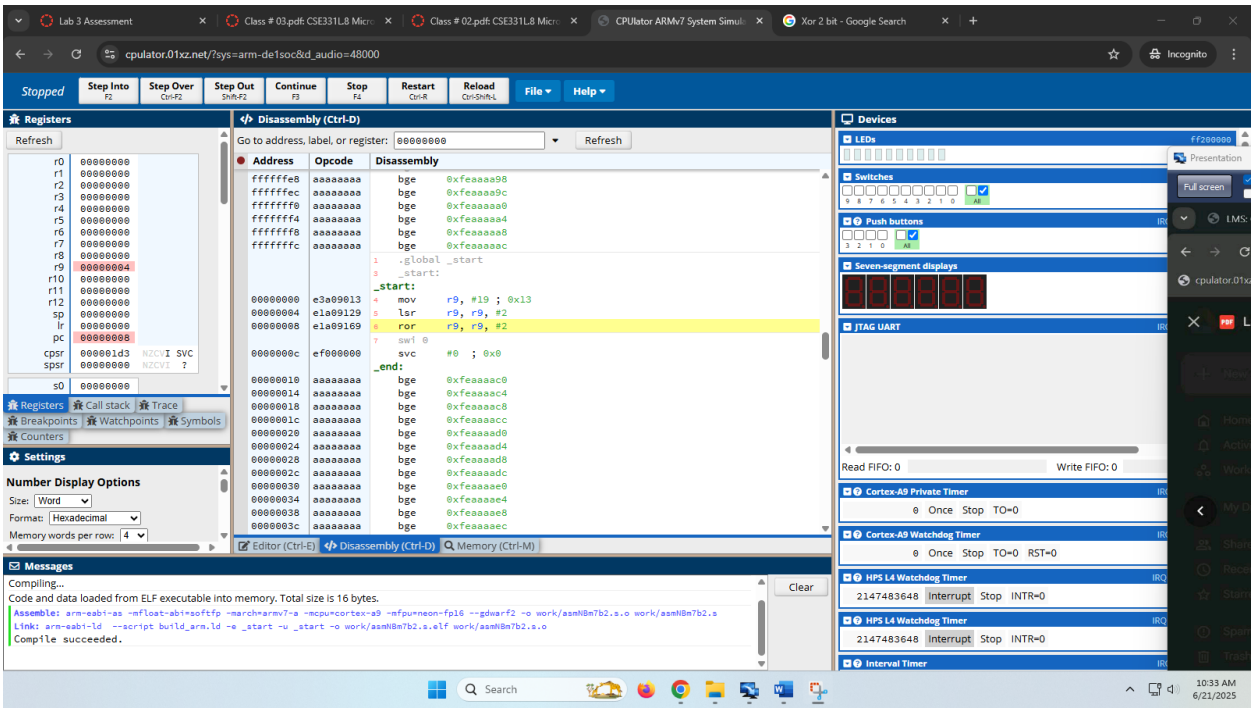
step over 1: 19 = 00000000000000000000000000010011 (32 bit)



step over 2: 19 = 00000000000000000000000000010011 (32 bit)

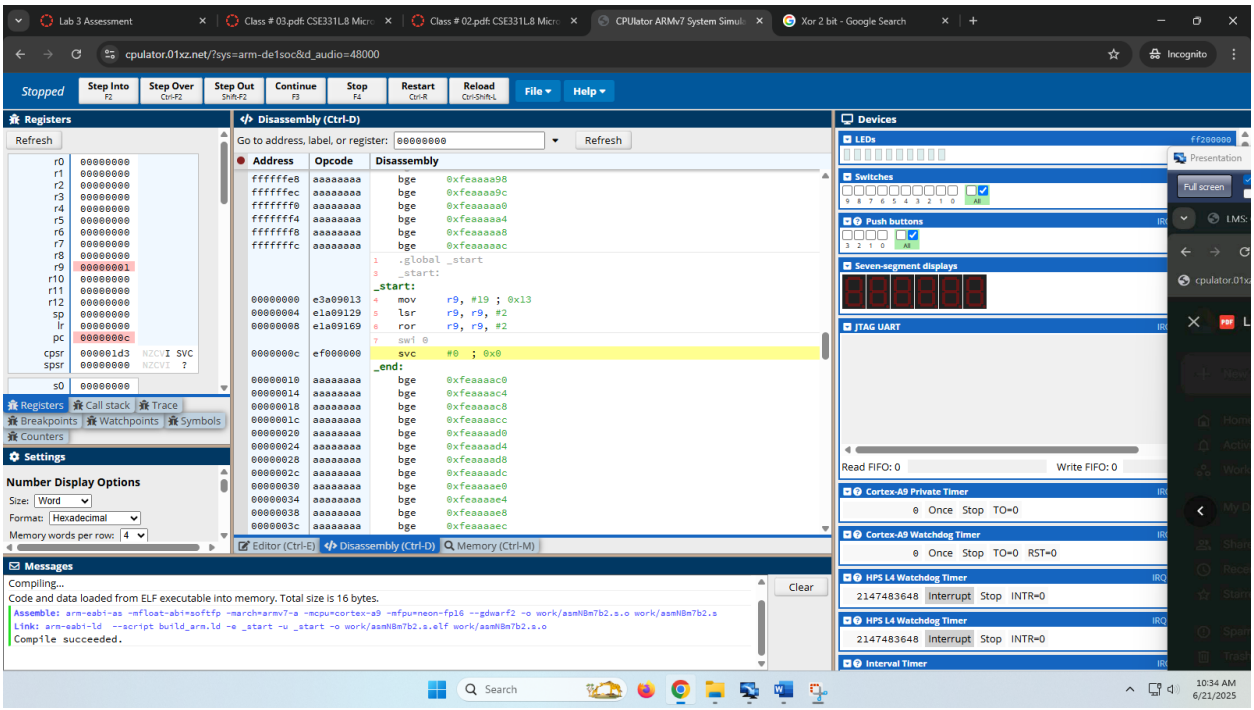
after right shift 2 we get

4 = 00000000000000000000000000000100 (32 bit)



step over 3: 4 = 00000000000000000000000000000100 (32 bit)

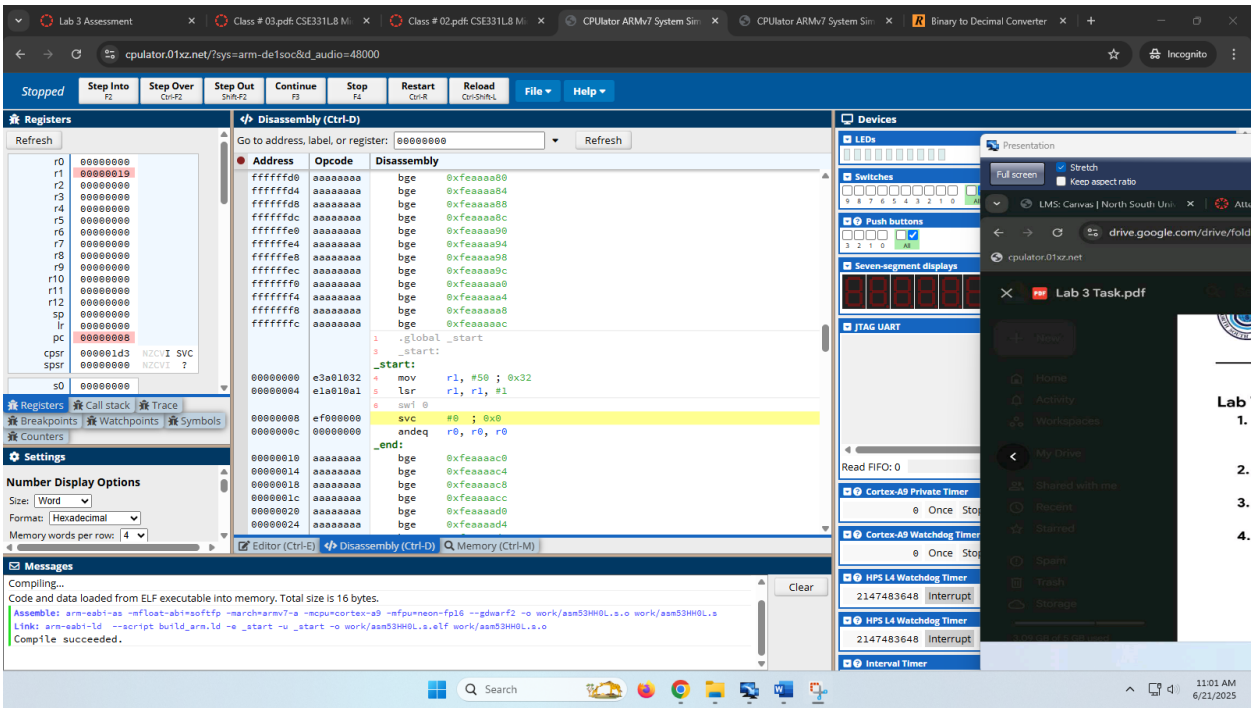
after ROR 2 we get , 1 = 00000000000000000000000000000001 (32 bit)



Q3

```
.global _start


_start:
        mov r1,#50
        lsr r1,r1,#1
        swi 0
```



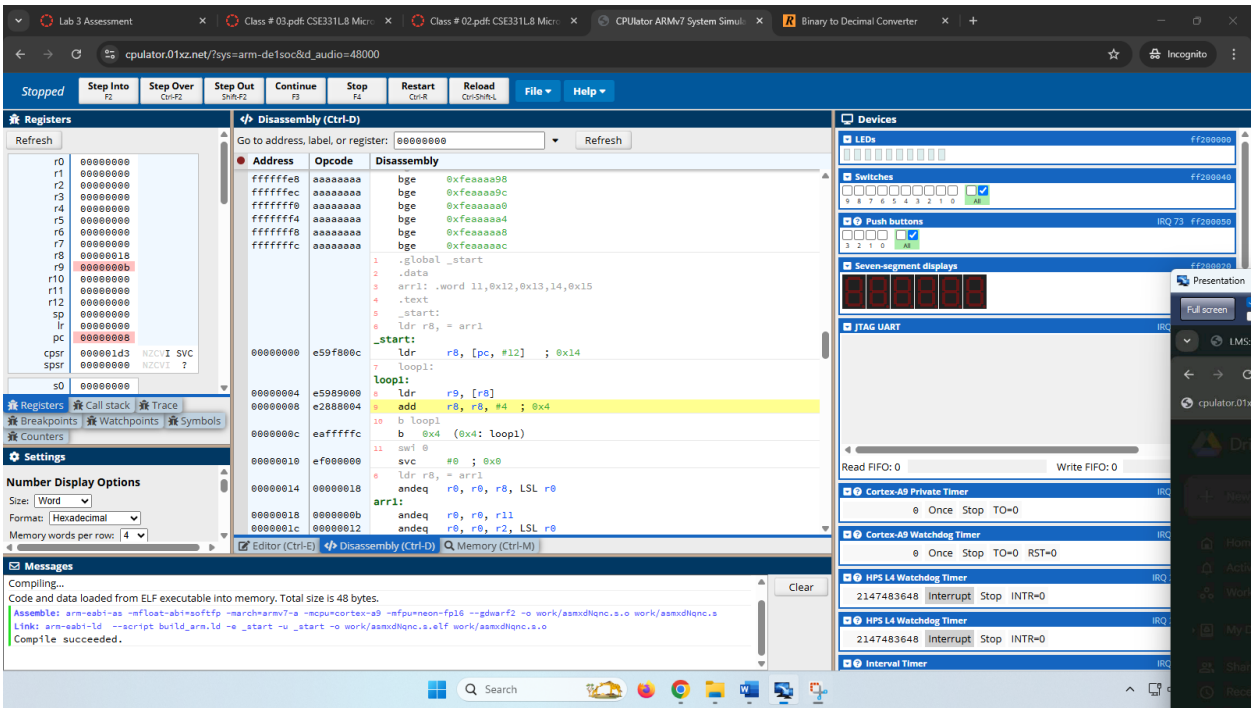
it said divide 2 so, Right Shift (LSR) will be 1 time

Q4

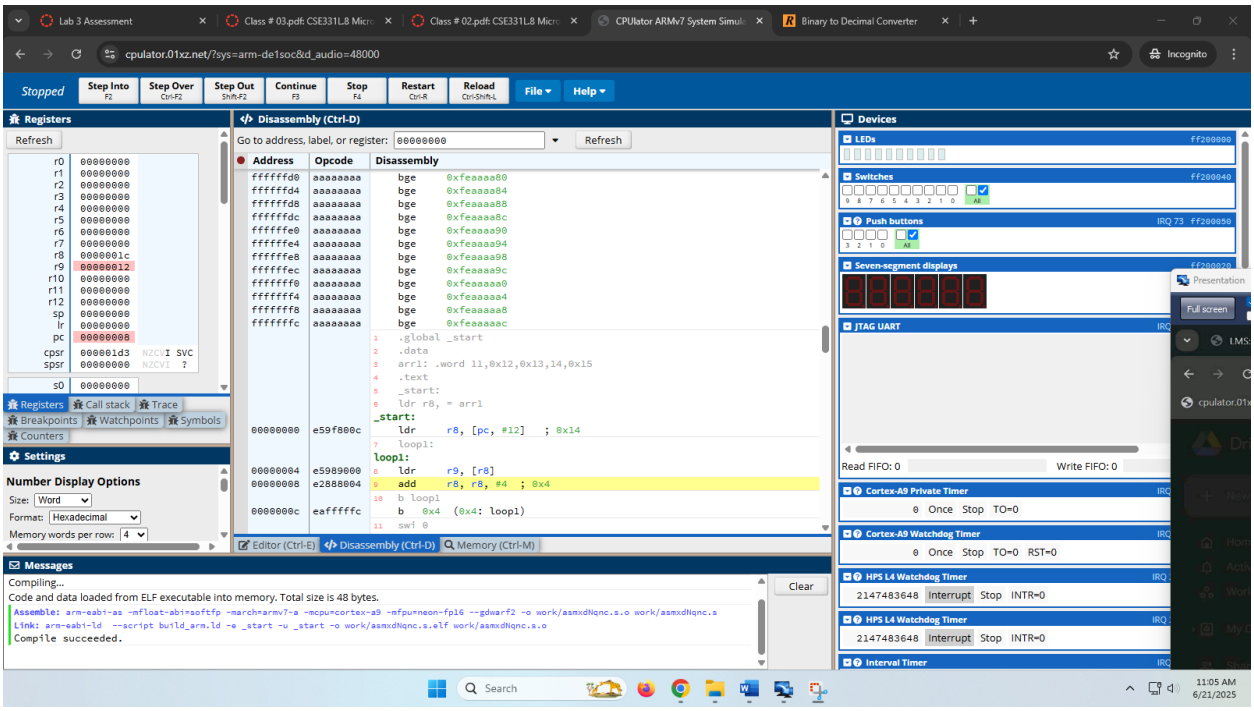
```
.global _start
.data
  arr1: .word 11,0x12,0x13,14,0x15
.text
```

```
_start:
        ldr r8, = arr1
        loop1:
        ldr r9,[r8]
        add r8,r8,#4
        b loop1
        swi 0
```

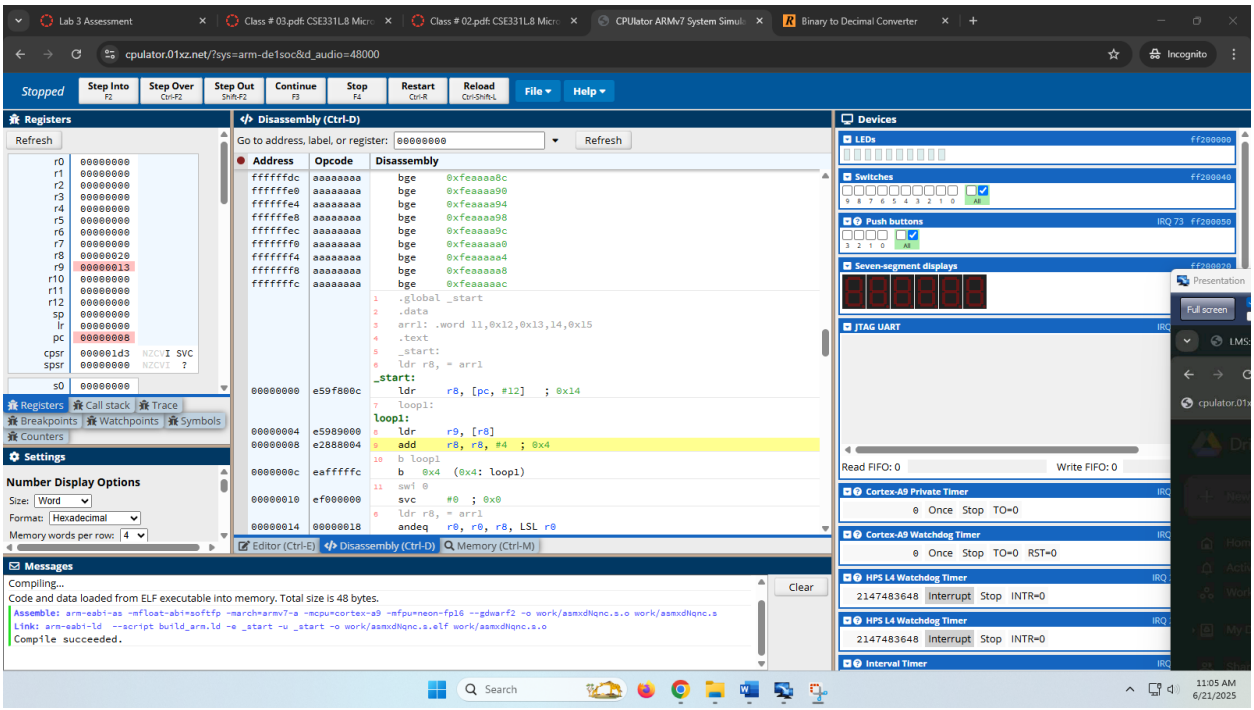
step over 1:



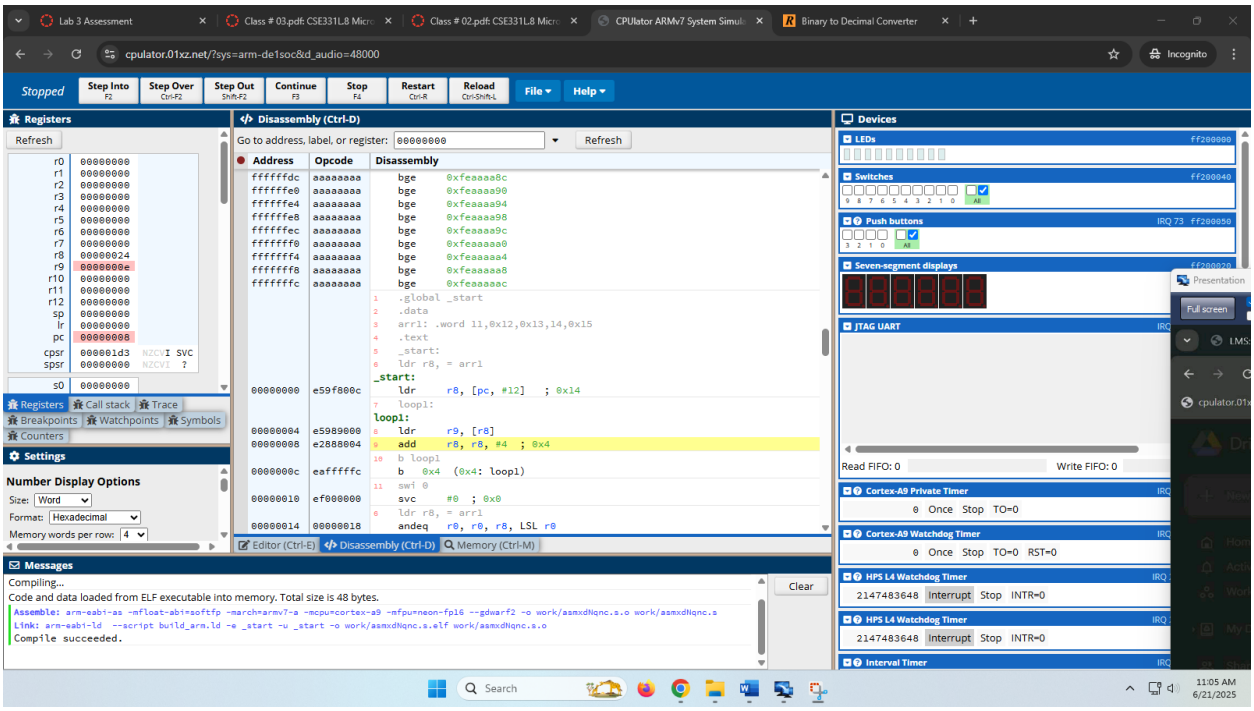
step over 2:



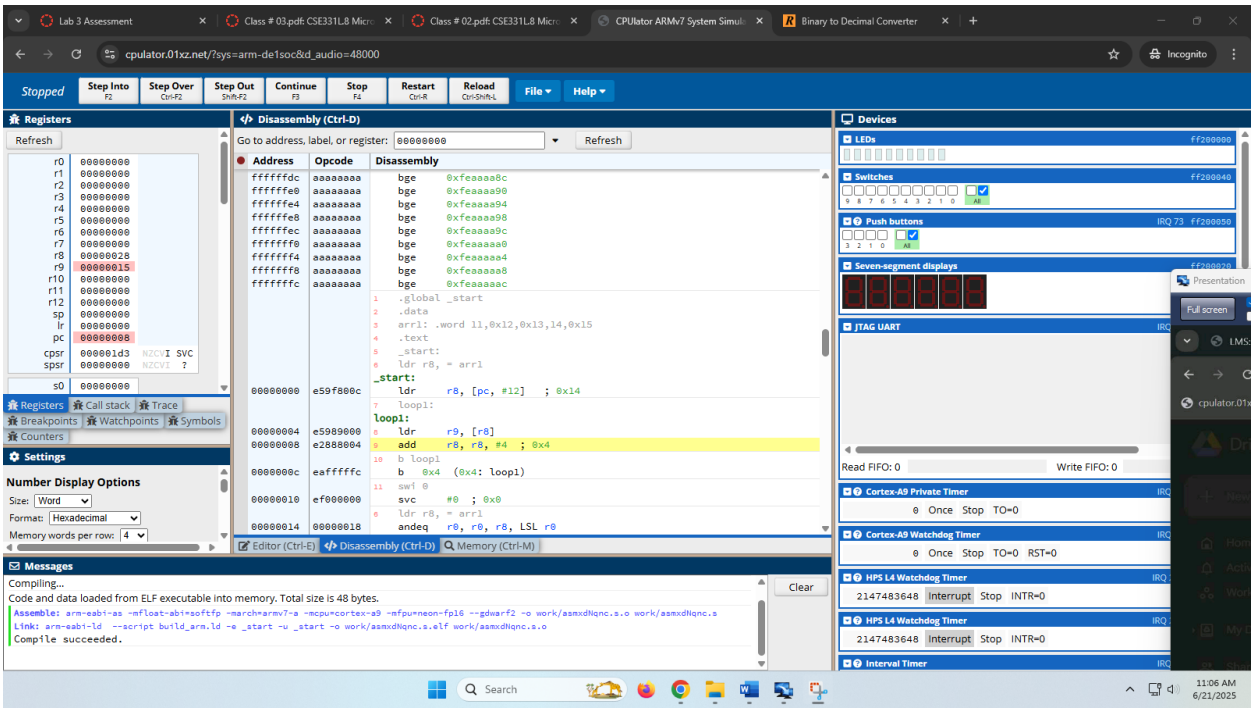
step over 3:



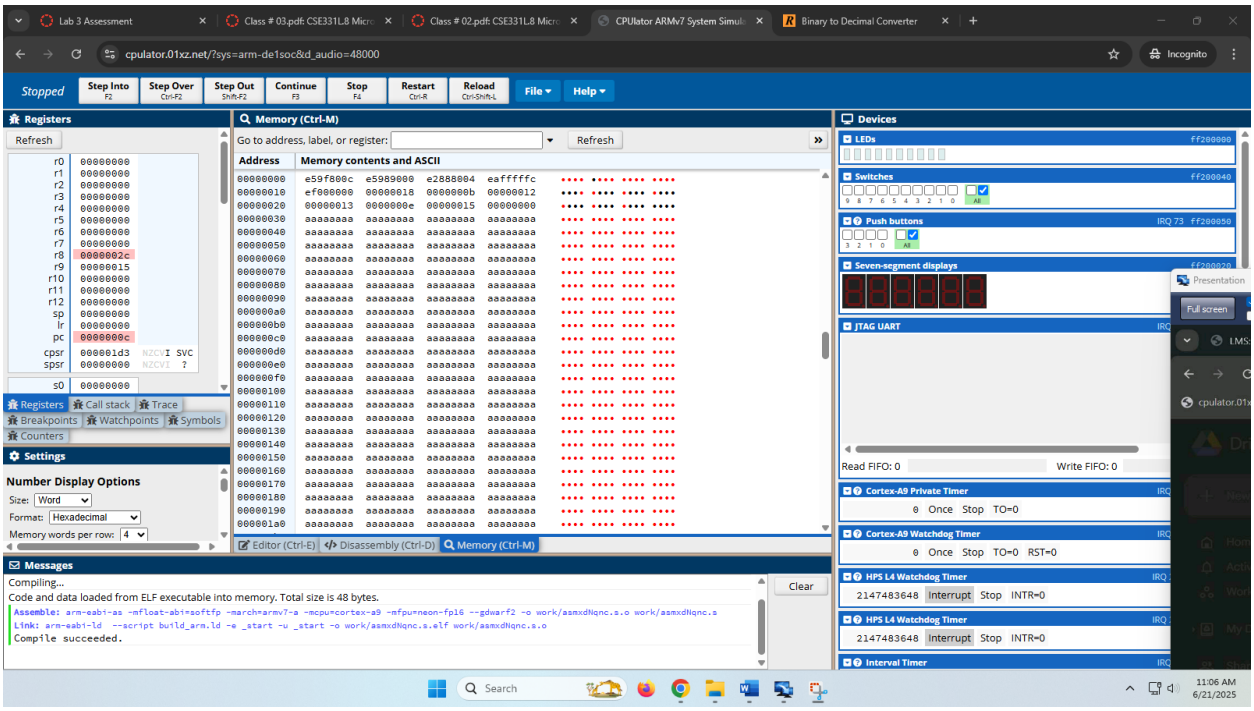
step over 4:



step over 5:



Memory:


Lab 3 Assessment
Class # 03.pdf: CSE331L8 Micro
Class # 02.pdf: CSE331L8 Micro
CPUlator ARMv7 System Simula
Binary to Decimal Converter
cpulator.01xz.net/?sys=arm-de1soc&d_audio=48000
Incognito
Stopped
Step Into F2
Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
File
Help
Registers
Refresh
r0 00000000
r1 00000000
r2 00000000
r3 00000000
r4 00000000
r5 00000000
r6 00000000
r7 00000000
r8 0000002c
r9 00000015
r10 00000000
r11 00000000
r12 00000000
sp 00000000
lr 00000000
pc 0000000c
cpsr 000001d3 NZCVI SVC
spsr 00000000 NZCVI ?
s0 00000000
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Memory (Ctrl-M)
Go to address, label, or register:
Refresh
Address
Memory contents and ASCII
00000000 e59f800c e5989000 e2888004 eafffffc
00000010 ef000000 00000018 0000000b 00000012
00000020 00000013 0000000e 00000015 00000000
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays ff200020
JTAG UART
Read FIFO: 0
Write FIFO: 0
Cortex-A9 Private Timer
0 Once Stop TO=0
Cortex-A9 Watchdog Timer
0 Once Stop TO=0 RST=0
HPS L4 Watchdog Timer
2147483648 Interrupt Stop INTR=0
HPS L4 Watchdog Timer
2147483648 Interrupt Stop INTR=0
Interval Timer
Messages
Clear
Compiling...
Code and data loaded from ELF executable into memory. Total size is 48 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmxdNqnc.s.o work/asmxdNqnc.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmxdNqnc.s.elf work/asmxdNqnc.s.o
Compile succeeded.
Presentation
Full screen
Search
11:06 AM
6/21/2025